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Artem Makoian

Programming: Python 3, C++
Languages: English (C1), Russian (native)
github.com/MakArtKar
scholar.google.com
leetcode.com/MakArtKar

Skills

Libraries	PyTorch, JAX, Transformers, Hydra, DVC, WandB, OpenCV, Sklearn, Numpy, Pandas, ONNX, Seaborn
Tools	SLURM (DDP, FSDP), SQL, Linux, Bash, Git, LaTeX, MarkDown

Technical Experience

JetBrains **Feb 2025 — Present**
ML Researcher / JetBrains Research ***with D. Vetrov*** *Python 3, PyTorch, transformers, Hydra, Lightning, SLURM*

→Masked Diffusion LMs
- Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
- G-Star: **ICML 2026** main track; multi-node training; PyTorch

→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference
- Distillation (flow maps, consistency) and latent space compression for real-time acceleration

Yandex **Jun 2023 — Sep 2024**
Middle ML Engineer / Geo services *Python 3, LangChain, PyTorch, Hydra, OpenCV, YT, Shapely, YQL, Nirvana*

→Road segmentation from aerial imagery with IoU 0.93
- Roads post-processing algorithms - reduced number of artifacts from $10\% \to 0.02\%$ and improved visual quality.
- Safety islands segmentation and vectorization with IoU 0.37

→News classification to create map updates hypotheses. Precision: 0.9, recall: 0.85.
- Developing and maintenance of the news pipeline: site parsing (with LangChain AI agents), structuring data and classifying.

Sber Devices **Jul 2022 — Jun 2023**
Middle ML Engineer / Computer Vision Platform *Python 3, C++, PyTorch, Hydra, DVC, OpenCV, ONNX, torch_pruning*

→Background removal for video conferencing with 0.98 Dice score
- Improved human segmentation quality working on segmentation pipeline, pruned model getting 1.5x speed acceleration
- Improved visual quality with post-processing algorithms: aligning the lighting and smoothing out the border.

→Trained a 700 KB model for payment card border segmentation and developed C++ inference pipeline in multiplatform mobile SDK

AIRI (Artificial Intelligence Research Institute) **Jul 2022 — Oct 2022**
ML Research Intern / GAN adaptation *Python 3, PyTorch, Hydra, WandB*
- Few-Shot Image Generation implementation for few-shot GAN adaptation to a new domain from ∼10 images

Yandex **Jul 2021 — Oct 2021**
ML Researcher / VR Team ***with V. Lempitsky*** *Python 3, PyTorch, Tensorboard*

→Trained model that predicts human's alpha mask to improve the quality of a VR scene
- Prepared datasets with human alpha masks, read and compared papers on an alpha matting problem
- Trained an improved FBA Matting net without using trimaps

Samsung R&D Department **Nov 2020 — Jun 2021**
ML Research Intern / Mobile Authentication *Python 3, PyTorch, Bash, Numpy, Pandas, Sklearn, Tslearn*

→Worked on a pipeline for mobile phone authentication by user's motion patterns with $82.2\% \pm 9\%$ accuracy - paper.
- Collected, preprocessed and handled data from devices. Used ML and DL models for authentication, got 70% accuracy baseline

Education

PhD in Computer Science (1st year), *Constructor University, Bremen, Germany* Sep 2026 — Present
- Teaching Assistant (seminars): Deep Bayesian Models (Spring 2026), Bayesian Methods (Fall 2026)
- Co-supervising 3 MSc theses, including Loop Transformers and Gaussian Diffusion LMs

Master in Computer Science, *Constructor University, Bremen, Germany* Sep 2024 — May 2026
Bachelor in Computer Science, *Neapolis University of Paphos, Cyprus*, GPA 86/100 Jan 2023 — Jun 2024
Bachelor in Computer Science, *HSE University, Moscow, Russia*, GPA 8.7/10 Sep 2019 — Jun 2024
Master's Equivalent in Data Science, *Yandex School of Data Analysis, Russia* Sep 2020 — May 2022

Selected courses: Machine Learning, Deep Learning, Generative Models, Bayesian Methods, Optimization Methods, Large-Scale DL, NLP, Computer Vision, Deep Vision, 3D Computer Vision, Reinforcement Learning, Probability Theory, Statistics, Linear Algebra, Matrix Computations, Algorithms and DS